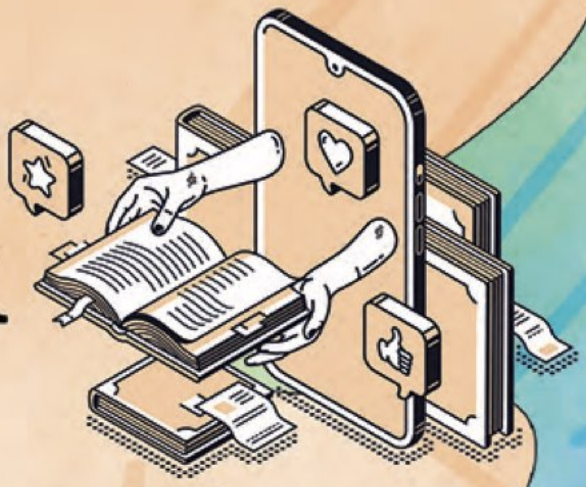
Education and New Developments 2021



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FOREWORD

This book contains the full text of papers and posters presented at the International Conference on Education and New Developments (END 2021), organized by the World Institute for Advanced Research and Science (W.I.A.R.S.), that this year had to be transformed into a fully Virtual Conference as a result of the Coronavirus (COVID 19) pandemic.

Education, in our contemporary world, is a right since we are born. Every experience has a formative effect on the constitution of the human being, in the way one thinks, feels and acts. One of the most important contributions resides in what and how we learn through the improvement of educational processes, both in formal and informal settings.

The International Conference seeks to provide some answers and explore the processes, actions, challenges and outcomes of learning, teaching and human development. The goal is to offer a worldwide connection between teachers, students, researchers and lecturers, from a wide range of academic fields, interested in exploring and giving their contribution in educational issues. We take pride in having been able to connect and bring together academics, scholars, practitioners and others interested in a field that is fertile in new perspectives, ideas and knowledge.

We counted on an extensive variety of contributors and presenters, which can supplement our view of the human essence and behavior, showing the impact of their different personal, academic and cultural experiences. This is, certainly, one of the reasons we have many nationalities and cultures represented, inspiring multi-disciplinary collaborative links, fomenting intellectual encounter and development.

END 2021 received 478 submissions, from more than 40 different countries, reviewed by a double-blind process. Submissions were prepared to take form of Oral Presentations, Posters and Workshops. The conference accepted for presentation 160 submissions (34% acceptance rate), from which, 151 submissions are published in full text in this book.

The conference also includes:
- One Keynote presentation by Dr. Hava Vidergor (PhD, senior lecturer of curriculum and instruction in the Graduate School at Gordon Academic College of Education, Israel).
- One Special Talk by Prof. Dr. Nazario Zambaldi (PhD General Pedagogy, Social Pedagogy, General Education and Subject Didactics at the Free University of Bolzano/Bozen, Italy)
We would like to express our gratitude to our invitees.

This conference addressed different categories inside the Education area and papers are expected to fit broadly into one of the named themes and sub-themes. To develop the conference program, we have chosen four main broad-ranging categories, which also covers different interest areas:

• In **TEACHERS AND STUDENTS**: Teachers and Staff training and education; Educational quality and standards; *Curriculum* and Pedagogy; Vocational education and Counselling; Ubiquitous and lifelong learning; Training programs and professional guidance; Teaching and learning relationship; Student affairs (learning, experiences and diversity; Extra-curricular activities; Assessment and measurements in Education.

• In **PROJECTS AND TRENDS**: Pedagogic innovations; Challenges and transformations in Education; Technology in teaching and learning; Distance Education and eLearning; Global and sustainable developments for Education; New learning and teaching models; Multicultural and (inter)cultural communications; Inclusive and Special Education; Rural and indigenous Education; Educational projects.

• In **TEACHING AND LEARNING**: Critical, Thinking; Educational foundations; Research and development methodologies; Early childhood and Primary Education; Secondary Education; Higher Education; Science and technology Education; Literacy, languages and Linguistics (TESL/TEFL); Health Education; Religious Education; Sports Education.

• In **ORGANIZATIONAL ISSUES**: Educational policy and leadership; Human Resources development; Educational environment; Business, Administration, and Management in Education; Economics in Education; Institutional accreditations and rankings; International Education and Exchange programs; Equity, social justice and social change; Ethics and values; Organizational learning and change, Corporate Education.

This book contains the results of the research and developments conducted by authors who focused on what they are passionate about: to promote growth in research methods intimately related to teaching, learning and applications in Education nowadays. It includes an extensive variety of contributors and presenters, who will extend our view in exploring and giving their contribution in educational issues, by sharing with us their different personal, academic and cultural experiences.

We would like to express thanks to all the authors and participants, the members of the academic scientific committee, and of course, to our organizing and administration team for making and putting this conference together.

Hoping to continue the collaboration in the future,

Respectfully,

Mafalda Carmo
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Conference and Program Chair

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ANALYSING LESSON-BASED INTERVIEWS USING THE LESSON ACTIVITIES MAP (LAMAP) AS A VISUAL TOOL

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Abstract

During training as generalists, some teachers find it complex and challenging to teach songs and lead class singing. The Song Leading research project longitudinally examines case studies of 16 trainees to explore how they acquire and develop the knowledge and skills to conduct a class singing lesson. The data corpus consists of video-recorded lessons, audio-recorded lesson-based interviews and personalised open-ended questionnaires. In this paper some phases of the interview analysis are presented. As each interview is conducted while watching the video-recorded lesson, its analysis should not be separated from the lesson content. The central question guiding this analysis is: How can the analysis of a lesson-based interview be combined with the video analysis of the lesson itself? We present a visual system to combine the analysis of interviews and videos. This system involves the use of the Lesson Activities Map (LAMap) – the transcript of the class singing lesson – based on the methodology developed in the Song Leading project. During the thematic analysis of the interviews, the LAMap is a visual tool that allows the researchers to systematically describe the lesson moments that were the starting points of the teachers' reflections. In addition, LAMap is a visual tool for collecting initial codes and identifying relationships between potential interview themes. The implications of the use of a visualisation system for lesson-based interview analysis are an added value for the coherence of case study interpretation. The paper contributes to research in education by providing concrete examples of how to make a qualitative analysis process explicit.

Keywords: *Lesson activities map, reflexive thematic analysis, professional development, song teaching, song leading.*

1. Introduction

This paper presents some stages of the case study analysis of the Song Leading research project. The project focuses on the cultural practice of group singing in the school context. In many pre-schools and primary schools, the teaching of songs and the conducting of class singing are delivered by generalist teachers (de Vries, 2015). They are expected to achieve and/or consolidate subject-specific skills during their training (Russell-Bowie, 2009). The study aims to explore how pre-service generalist teachers carry out the task of song teaching and reflect on their song leading experiences at three different points during their professional training – initial, intermediate and final. We want to analyse how teachers experience their development, both from a personal perspective and from that of familiarisation and deepening of subject-specific knowledge and skills. The data corpus of the project consists of videos, interviews, and personalised open-ended questionnaires. In this paper some phases of the interview analysis are presented. The examples and discussion concern how the Lesson Activities Map (LAMap) (Savona et al., 2021) – i.e., the transcript of a song leading lesson created with the methodology developed in the Song Leading project – can be a tool for researchers to analyse the lesson-based interviews. The LAMap methodology contributes to qualitative research in education by providing a visual analytical tool. It enhances the characteristics of the subject matter, facilitates the systematic organisation of data, and supports the coherence of the interpretation of verbal data relating to a specific educational event and context.

2. Design

The Song Leading research project examines case studies of 16 pre-service teachers attending their three-year training course as generalists. The data set for each case study consists of three video-recorded lessons, three interviews – lesson-based and audio-recorded – and a personalised

open-ended questionnaire. This data was collected once a year during the pre-service teachers' internships. Five PhD students and two supervisors work on the ongoing project (2018-2022). Each PhD student has a background as a professional musician and singing or instrument teacher, and/or as a music educator at primary, secondary or tertiary level. The two supervisors are professors with research backgrounds in developmental and cultural psychology, arts education, and research methodology. This information is necessary for positioning the team members in the data analysis process.

3. Objectives

This paper aims to show how we use LAMap while analysing interviews. For the analysis of the interviews, we follow the reflexive thematic analysis as defined by Braun and Clarke (2006). They outline six phases for this process. For phases 2, 3 and 4 – *generating initial codes* (2), *searching for themes* (3) and *reviewing themes* (4) – Braun and Clarke (2006) suggest the creation of tables and mind-maps. This suggestion coincides with the LAMap as our visualisation system of a song leading lesson. Through examples, this paper demonstrates how LAMap is a useful visual tool for analysing lesson-based interviews.

4. Method

Videos, interviews and personalised questionnaires are analysed with different qualitative methods and methodologies. The interviews, on which we focus in this paper, are analysed through reflexive thematic analysis (Braun & Clarke, 2006) to identify and describe patterns in the way teachers relate their teaching experience with comments. The analysis is framed within the epistemological approach characterised as the teacher-object-learner paradigm called “didactics” (Reusser, 2008; Schneuwly, 2021). The identification of themes is guided by a semantic and explicit interpretation level based on the relationship between experience, meaning, and language (Wooffitt & Holt, 2011). Through reflexive thematic analysis we aim to report a detailed and subject-related description of each interview. In the project, each PhD student is responsible for transcribing video and interview data and guiding the analysis of several case studies. AS, the author of this paper, leads the analysis of Ruth's case, which serves here as an example to show the application of reflexive thematic analysis to lesson-based interviews. Each team member is familiar with the interviews from all the case studies. The analysis of each case is regularly discussed by the team at all stages. As each interview is conducted by watching the video-recorded lesson, the pre-service teacher's comments and reflections are often "triggered" by the context of the lesson in question. In the following paragraphs we provide examples of how the LAMap is used to produce the map of initial codes and potential themes generated by the interview interpretation. Indeed, 'the most basic segment, or element, of the raw data or information that can be assessed in a meaningful way regarding the phenomenon' (Boyatzis, 1998: 63) is often the lesson activity which can be visualised with the LAMap.

4.1. The LAMap as a visual domain summary

The LAMap was devised in the initial stage of the Song Leading project in an iterative manner in order to gain a systematic overview of the relevant and recurring activities of each lesson. The result is a methodology for transcribing class singing lessons. It consists of symbols and icons that are used to represent the course of recurrent activities that constitute lessons in this specific subject, such as making semantic gestures to learn the lyrics of the song. Although the analysis of the videos is guided by the same epistemological decisions as the analysis of the interviews, i.e., teacher-object-learner interaction (Schneuwly, 2021), the result of the video analysis is a domain summary (Braun & Clarke, 2019) and not themes generated by a reflexive thematic analysis. However, the combination of the two analyses contributes to an overview of how the content of the interviews refer to the actions during the lessons. Thus, the pre-service teachers’ comments can be directly linked to the course of actions during the lesson as visualised by the LAMap.

Figure 1 shows on the left side the LAMap of Ruth’s first leading a class singing lesson. The LAMap is divided into two parts by the speaking line, along which the spoken communication between teacher and children is marked. Ruth's actions are written above the speaking line and the children's below. On the right, the Song Leading Activities Key provides definitions for the symbols and icons that represent the song leading activities in LAMap.

4.2. The LAMap as a tool for reflexive thematic analysis in education

The following figures were created for analysing Ruth's first leading of a class singing lesson. Note that they do not show the final results of the analysis, but rather represent the phases 2, 3 and 4 as outlined by Braun and Clarke (2006). Thereby, the LAMap serves as a graphical tool for:

1) Systematically locating the initial codes of the interview sections during the course of the lesson (phase 2, figure 2 A).
2) Contextualising interview extracts concerning the same code to check their coherence before 'nominating' them for potential themes (phase 3, figure 2 B).
3) Creating a mind-map of the initial themes generated by the analysis (phase 4, figure 3).

Figure 2 (A) shows the LAMap of Ruth's lesson. The vertical lines on the LAMap indicate the moments when Ruth paused the recording to comment. The interview started with an opening dialogue (OD), the video-viewing sections are numbered consecutively (in the figure: 1, 2 and 3), and a concluding dialogue (CD) took place at the end of the video-viewing. The lower part shows the summary table of the initial codes. The code 'known song vs. new song' is highlighted in grey and is selected for the next analysis phase. Figure 2 (B) shows the extracts related to the 'known song vs. new song' code. It provides information about the team member responsible for transcription and translation, the interview section (OD, 1, 2, 3, CD) and the location of the extracts in the transcript (text chunks). Screening the extracts for a single theme is a necessary step to check the coherence of the coding and to nominate initial codes for potential themes.

Figure 3 shows the initial thematic map of the lesson-based interview combined with the LAMap. The symbols and icons highlighted in grey indicate the lesson moments on which Ruth commented while watching the video. From these extracts initial themes and sub-themes were generated, the interpretation of which requires completion through the description of those lesson moments. Other initial themes have been generated from reflections of a general nature and not strictly related to the lesson activities.

5. Discussion

The development of visual tools like the ones presented in this article is desirable when analysing interviews based on a context “bound” to a specific event. In the case of the Song Leading project this is the lesson-based interview. Consequently, part of the interview content is derived from the input the interviewer receives while watching the video-recorded lesson. During the analysis of the interviews, the LAMap is a tool for the researchers to systematically locate and describe the lesson moments that gave rise to the teacher's comments. By taking into account the teaching content enacted during the lesson and graphically transcribed in the LAMap, we achieve a convincing coherence between the observed actions and generated themes. However, this essential aspect of analysis remains still a challenge. The transition from description to abstraction of themes with shared meanings requires both familiarity and a distancing from the lesson content (not the interviews!). Researchers have to go beyond description and produce themes with generalised meanings in order to reconstruct and understand how teachers develop skills and how they conceive of their own professional development.

6. Conclusion

The aim of this paper was to show how the transcription of a lesson can support lesson-based interview analysis. One example is the use of the LAMap – the transcription of a class singing lesson with the methodology developed in the Song Leading project. The visualisation of the transcribed lesson combined with the interview analysis can offer new possibilities for qualitative research into subject-specific teaching. Comprehensive analytical narratives can be created through both the systematic description of teachers' subject-related actions and the coherent interpretation of their personal perspectives of their own professional development.

Figure 1. LAMap of the first song leading lesson of the pre-service teacher Ruth.

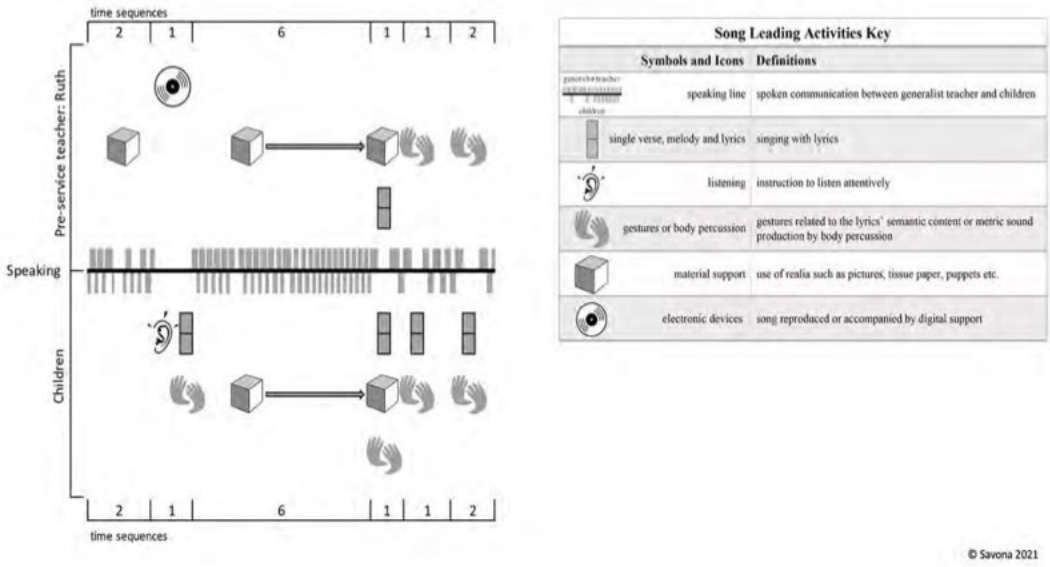


Figure 2. LAMap and map of initial interview codes (A). AMap and data extracts coded for 'known song vs. new song' (B).

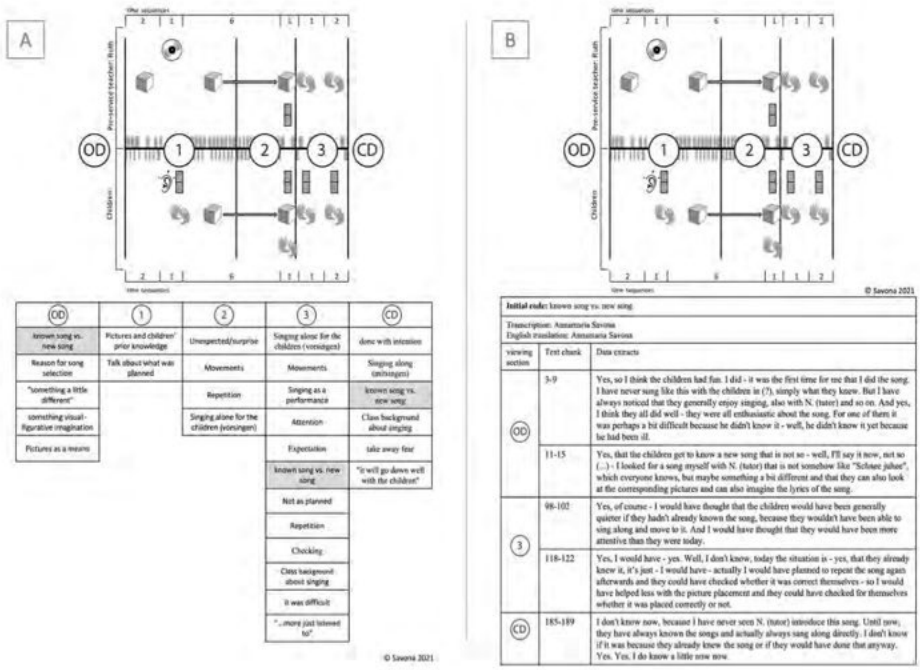
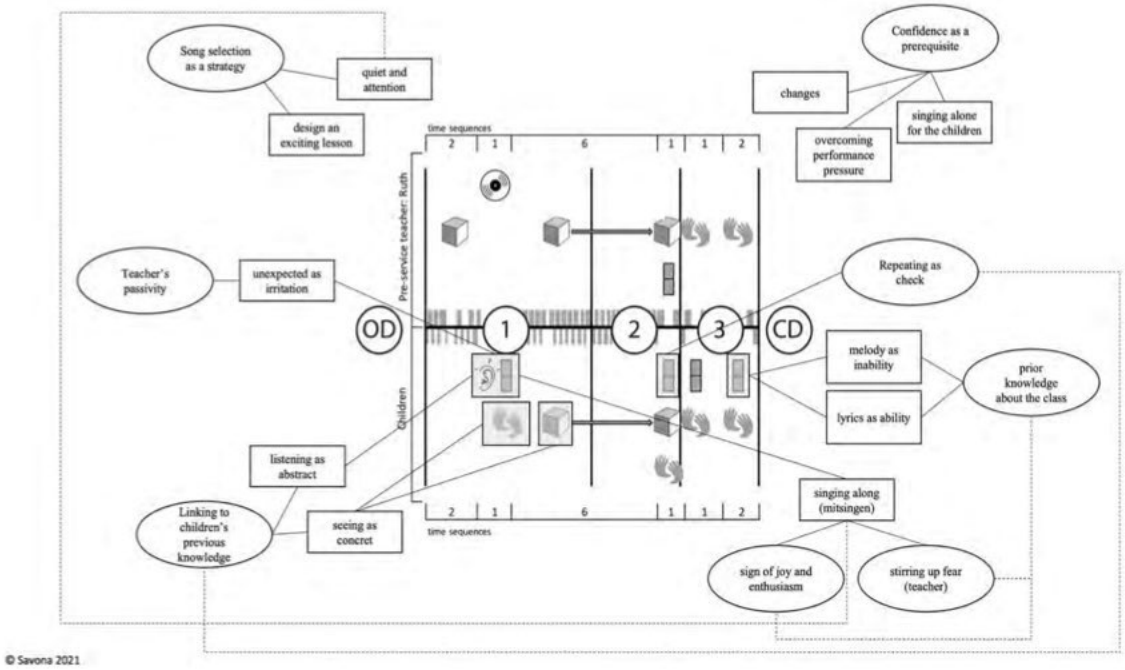


Figure 3. Initial thematic map of Ruth's first lesson using LAMap.



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